

Noken Descent

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Abstract

A Noken implicit model is used to jointly learn query, key and value embeddings. In this context, a query update refines query-key alignment by lowering entropy. On this systematic foundation, we then learn layers of feed forward networks to polish perplexity, resulting in a coherent attention mechanism.

Noken Descent

Given token string $t_1, \dots, t_m \in \mathbb{R}^r$

learn matrices $K_{n \times n}$, $Q_{n \times r}$, $V_{n \times n}$, $Q'_{n \times r}$

and fixed points $s_i \in \mathbb{R}^n$, $n > r$,

a latent token, we call a Noken,

so that in the following implicit model [1]:

$$s_i = \varphi(Ks_i + Qt_i)$$

$$0 = Vs_i - Q't_i$$

where $0 \in \mathbb{R}^n$ and $i \in [m]$,

$$K = \theta - \text{diag}(\theta \mathbb{1}_n) + \delta \mathbb{I}_n, \quad 0 < \delta < 1,$$

$\theta \in \mathbb{R}^{n \times n}$, $\mathbb{1}_n = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}_{n \times 1}$ and $\mathbb{I}_n \in \mathbb{R}^{n \times n}$ the identity matrix,

ensures [2] K is well-posed for φ , that is

Fixed points s_i exist and are unique.

Let $k_i = Ks_i$, $q_i = Qt_i$, $v_i = Vs_i$ and $q'_i = Q't_i$.

Note that $q'_i = v_i$, $i \in [m]$.

Define $a_{ij} = \langle q'_i, k_j \rangle$ and $h_i = \sum_{j=1}^m a_{ij} v_j$.

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For $i \in [m]$, define update $\frac{1}{q}'_i = v_i - h_i$,

$\frac{1}{a}_{ij} = \langle \frac{1}{q}'_i, k_j \rangle$ and $\frac{1}{h}_i = \sum_{j=1}^m \frac{1}{a}_{ij} v_j$.

$$\frac{1}{a}_{ij} = \langle \frac{1}{q}'_i, k_j \rangle = a_{ij} - \langle h_i, k_j \rangle$$

$$\Rightarrow \frac{1}{\alpha}_{ij} \triangleq \text{softmax}_j \frac{1}{a}_{ij}$$

$$= e^{\frac{1}{a}_{ij}} / \Sigma', \quad \Sigma' \text{ a normalizer}$$

$$= (e^{a_{ij}} / \Sigma) (\Sigma / e^{\langle h_i, k_j \rangle} \Sigma'), \quad \Sigma \text{ a normalizer}$$

$$= \alpha_{ij} c_{ij}$$

$$\frac{1}{\alpha}_{ij} = c_{ij} \alpha_{ij} \Rightarrow H(\{\frac{1}{\alpha}_{ij}\}) \leq H(\{\alpha_{ij}\})$$

Hence query updating (once) lowers entropy. (cf [3])

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Given h_i , define $y_i = \text{FFN}_\gamma \left(\sum_{j=1}^m \frac{1}{\alpha_{ij}} v_j \right)$, $i \in [m]$

where $\frac{1}{\alpha_{ij}} = \text{softmax}_j \frac{1}{a_{ij}}$ and FFN_γ is a

Feed forward network with parameters γ .

Define probabilities $\beta'_{ij} = \text{softmax}_j \text{score}(y_i, k_j)$, $i \in [m]$

where score is a desired similarity function eg a kernel,

and define $j_i^{\beta'} = \arg \max_{j \in [m]} \beta'_{ij}$.

Given input token t_i , define output $o_{i+1} = t_{j_i^{\beta'}}$

and learn γ to minimize perplexity $(o_2, \dots, o_{m+1})$.

This is the first attention layer. (cf [4])

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Define attention update $y'_i = \text{FFN}_{\gamma'} \left(\sum_{j=1}^m \beta'_{ij} v_j \right)$

and let $\beta''_{ij} = \text{softmax}_{j \in [m]} \text{score}(y'_i, k_j)$, $i \in [m]$

and $j_i^{\beta''} = \arg \max_{j \in [m]} \beta''_{ij}$, $o'_{i+1} = t_{j_i^{\beta''}}$ as before.

Learn γ' to minimize perplexity $(o'_2, \dots, o'_{m+1})$.

As with transformer attention repeat these layers.

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Summary

$\frac{1}{q_i}$ has distribution $\frac{1}{\alpha_{ij}}$ over $\{k_j\}$ and query updating

lowers entropy. y'_i has distribution β''_{ij} over $\{k_j\}$

and attention updating polishes perplexity. From a Noken

implicit model we jointly learn Q-K-V embeddings, update

queries to lower entropy, and layer Feed Forward networks

to polish perplexity.

□

References

[1] 'Implicit Deep Learning'

Laurent El Ghaoui, Fangda Gu, Bertrand Travacca,

Armin Askari, Alicia Y. Tsai

arXiv:1908.06315v4 [cs.LG] 6 Aug 2020

[2] 'Robust Implicit Networks via Non-Euclidean Contractions'

Saber Jafarpour, Alexander Davydov,

Anton V. Proskurnikov and Francesco Bullo

arXiv:2106.03194v6 [cs.LG] 26 Jan 2022

[3] 'Noken Rademacher Descent'

Gary Nan Tie, Jan 20, 2026

DOI: 10.13140/RG.2.2.20368.47367

[4] 'Imformer: an implicit model of attention'

Gary Nan Tie, Mar 20, 2026

DOI: 10.13140/RG.2.2.11585.83041